

Certificate H and the Klein Stand

the Gross Gamma-normalization, the volume 1125, and the heptad arithmetic

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 16/16

1 Setting

Certificate H — the summit of the program arch: the full Gross Γ -normalization on the $N = 15/30$ stand, closing the Γ -remainder of certificate G. The third stand — the Klein quartic Jacobian with its arithmetic — is assembled here as well.

Theorem 1 (H1 — the cyclotomic lattice). *The CM lattice: $\mathbb{Q}(\zeta_{30}) = \mathbb{Q}(\zeta_{15})$ — one stand, two Γ -levels. The Riemann form is integral (Ramanujan sums); the polarization type (SNF) is $(1, 1, 5, 5, 15, 15, 15, 15)$; the discriminant is $3^4 \cdot 5^6 = 1265625 = 1125^2$.*

Proof. The Ramanujan form is integral; the SNF is computed by the exact integer algorithm; the product of invariant factors $1 \cdot 1 \cdot 5 \cdot 5 \cdot 15^4 = 1265625$ matches $3^4 \cdot 5^6$ — verified by factorization. The field: $\phi(30) = \phi(15) = 8$; the isomorphism $\zeta_{30} \mapsto -\zeta_{15}^8$. $\square$

Theorem 2 (H2 — the Γ -volume).

$$\text{vol}_h = (2\pi)^{32} \prod_{k=1}^{N-1} \Gamma(k/N)^{-c_k} = 1125 \quad (N = 15 \text{ and } 30),$$

the exponents c_k exact (for $N = 15$: $c_k = 4$ at ten values of k , $c_k = 6$ at $k = 3, 6, 9, 12$); the residual $1.01 \cdot 10^{-119}$; $\det V^2 = 1265625/256$; PSLQ deflation ($N = 15$) recovers the integer exponent vector; $N = 30$ certified by the sine count.

Proof. The Chowla–Selberg product over characters: the exponents collect the conductor multiplicities h_d . Computation at $dps = 120$ gives 1125 with residual 10^{-119} ; the agreement with $\sqrt{\text{disc}}$ of H1 is exact. The sine count: $\Gamma(k/30)\Gamma((30-k)/30) = \pi / \sin(\pi k/30)$ closes the identity at even k without PSLQ. $\square$

Theorem 3 (H3–H6). (H3) *Every Fermat period is the Γ -monomial $\frac{1}{N}\zeta^{ra+sb}\Gamma(a/N)\Gamma(b/N)/\Gamma((a+b)/N)$ (91/406 periods; quadratures 10^{-121}); level 30 with odd k is not reducible to 15. (H4) *The moduli $|\omega(\tau_i)|^2$ are Γ -monomials; the phase $\psi = \frac{(1+3\sqrt{5})+i(\sqrt{15}-\sqrt{3})}{8} = R/|R|$ is recovered by LLL; $|R| = 3 - \sqrt{5}$ is a unit of $\mathbb{Q}(\sqrt{5})$. (H5) *The ladder $\pi/7, \pi/15, \pi/30$ — rungs of one μ_N -quantum ladder; the weights $4 \sin(\pi k/N)$ agree. (H6) *The denominator bound $Q_{\text{stand}} = 2N \max s_i = 480$; the F rationalization closes without a scan.****

Proof. (H3) — the closed period form + certificate B (69 tests, $3.9 \cdot 10^{-36}$) + deep quadratures (10^{-121}); non-reducibility: $\Gamma(1/30)$ is not reduced by the inventory to denominator 15. (H4) — LLL in $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$ recovers the exact coordinates; $R\bar{R} = (3 - \sqrt{5})^2$ exact. (H5) — the phases $\pi/7$ (Klein), $\pi/15, \pi/30$ are levels of the construction $\delta = \pi/N$; the weights agree with the Ramanujan tables. (H6) — $2 \cdot 30 \cdot 8 = 480$; all measurements inside the F1 uniqueness intervals. $\square$

Protocol data: Klein — the stand within H

Smoothness: $28(xyz)^3 = 0$; genus 3; $\Delta = -343 = -7^3$; $c_4 = 105$; $j = -3375 = -15^3$; the form $W^2 = v^3 - 35v - 98$; roots $7, \frac{-7 \pm \sqrt{-7}}{2}$; $\Omega = 1.9333117056 \dots$ (three schemes, agreement $1.4 \cdot 10^{-32}$); $\tau = (1 + \sqrt{-7})/2$; $j(\tau) = -3375$ from the period (57 digits, anchor $j(i) = 1728$); $\zeta + \zeta^2 + \zeta^4 = \tau - 1$; isotypy $(1, 1, 0, 1)$; $\lambda_1(\text{Heawood}) = \sqrt{2}$, multiplicity 6; the Fano code $C_7 + \mu_4$.

Summary of results: summary

Certificate H accepted (6/6): the cyclotomic lattice, the volume 1125 in the full Γ -normalization, the Γ -monomial periods, the Gross normalization, the ladder, the denominators — the arch of the program is closed. Runtime ~ 2 minutes.

Verification: reproduction

`python3 laboratory.py` $\rightarrow$ “Certificates $\rightarrow$ H”; “Stands $\rightarrow$ Klein”; Lean: `stand_disc`, `snf_product`, `klein_integers`.